

EBN111 SEMESTER TOETS/TEST 1 (15 MAART 2012)
ANTWOORDBLAD / ANSWER SHEET

Student se besonderhede/Student's details	
Van en voorletters: <i>Surname and initials:</i>	MEMO
Graadkursus: <i>Degree Course:</i>	Studentenommer: <i>Student number:</i>
Tel. nr.: <i>Tel no.:</i>	

Q1	22	Q2	23	Tot:	45 /45
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VRAAG 1 – QUESTION 1 (22)	
1.1	$W = -2\text{ J}$
1.2	$v = -10\text{ MV}$ $i = -10\text{ mA}$
1.3	$a) P_1 = -15\text{ W}$ $P_2 = 3\text{ W}$
1.4	$b) P_2 \text{ absorbs power}$
1.5	$a) \text{ Nodes: } 5$ $b) \text{ Branches: } 7$
1.6	$c) V_S = 4\text{ V}$ $d) V_R = -2\text{ V}$
1.7	$e) I_X = 0.5\text{ A}$
1.8	$a) v_0 = -10\text{ V}$ $b) i_2 = 0.667\text{ A}$
1.9	$\text{Circuit (1): } R_{eq} = 3.667\text{ }\Omega$ $\text{Circuit (2): } R_{eq} = 6\text{ }\Omega$

VRAAG 2 – QUESTION 2 (23)

2.1	$a = 3$	$b = -3$
	$c = -3$	$d = 3$
2.2	$a = 5$	$b = 5$
	$c = -5$	$d = 10$
2.3	$V_2 = 0\text{ V}$	$i_l = 1\text{ A}$
2.4	$a = 1$	$b = -1$
	$c = 2$	$d = 1$
2.5	$a = -1$	$b = 1$
	$c = 31$	$d = 2$
2.6	$\begin{bmatrix} 0.5 & -0.2 & -0.2 \\ -0.2 & 0.2 & 0 \\ -0.2 & 0 & 0.3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 15 \\ -5 \end{bmatrix}$	